

tier1_avd_hvf_smoke

tier1_avd_hvf_smoke.sh — AVD-on-HVF boot smoke (emulation-fabric P1 / T1 tier)

Field	Value
Revision	1
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Status	active

Overview

First deliverable of emulation-fabric **P1 (T1 tier)** per [../design/emulation_fabric/ROADMAP.md](#): prove the only hardware-accelerated Android path on the Apple-Silicon dev host (REPORT §2.2) is operational by booting an **arm64-v8a** Android AVD headless on Apple **Hypervisor (HVF)** and asserting a completed boot.

Prerequisites

- ANDROID_HOME Android SDK with emulator/ + platform-tools/adb.
- An installed **arm64-v8a** system image and an AVD (e.g. Pixel_8, API 35).
- macOS on Apple Silicon (HVF; there is no /dev/kvm — REPORT §0).

Usage

```
bash tests/emulator/tier1_avd_hvf_smoke.sh
AVD=Pixel_8 PORT=5584 BOOT_TIMEOUT=200 bash tests/emulator/
tier1_avd_hvf_smoke.sh
```

Anti-bluff contract (§11.4)

PASS requires BOTH: 1. `sys.boot_completed == 1` (the device actually finished booting), AND 2. `ro.product.cpu.abi == arm64-v8a` — the **acceleration**

proof: an x86/x86_64 abi on this host would mean slow TCG emulation, not HVF, so the tier would not be the accelerated path P1 requires.

Evidence (under `docs/qa/<run-id>-avd-hvf-smoke/`): the emulator boot log + `boot_smoke_result.txt` (`avd/serial/booted/abi/sdk` + accel log lines).

Edge cases / internal behaviour

- **emulator not on PATH:** the script uses the absolute `$ANDROID_HOME/emulator/emulator` (the non-interactive shell does NOT have emulator on PATH — the lesson from the first attempt; §11.4.6 captured as FACT).
- **Cleanup (§11.4.14 / §12):** ALWAYS tears down via `adb -s emulator-<port> emu kill` (serial-targeted, graceful — NEVER a broad `pskill qemu.*`, which could touch podman's qemu). The graceful shutdown takes ~20s and continues after the script exits; the emulator self-terminates (verified: no orphan, podman untouched).
- **Boot timeout:** `BOOT_TIMEOUT` (default 200s); a non-boot FAILs with the log path.

Related

- `docs/research/emulation_infra/REPORT.md` (§0 host fact, §2.2 AVD-HVF verdict)
- `docs/design/emulation_fabric/{DESIGN,ROADMAP,TEST_COVERAGE_PLAN}.md`
- Next P1 steps (NOT yet done): build/install the `ota-android-agent` harness APK on the booted AVD + drive its decision logic; resolve the **UNCONFIRMED:** GSI-A/B `real-update_engine` question (REPORT §2.2 / §8).

Last verified: 2026-06-10 (booted=1, abi=arm64-v8a, sdk=35; clean teardown).